Microbes & cells — overview

Immune cells, pathogens and body cells side by side

Variant: standard (~60–70 entries)

Contents

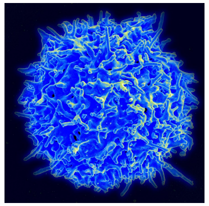
Where a real microscope photograph is available, it is shown; otherwise a stylised microscope-style sketch. The infographic and kids styles are hand-drawn SVGs.

Immune cells — Lymphocytes

The specialised troops of adaptive immunity.

Lymphocytes are small white blood cells with a large round nucleus. They mature in bone marrow (B, NK) or the thymus (T), circulate through blood and lymph, and gather in lymph nodes, spleen and mucosal tissues.

Helper T cell (CD4)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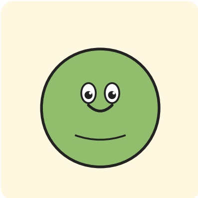
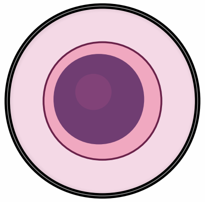

KIDS STYLE

FUNCTION: Conductor of the immune response: recognises antigen fragments shown by other cells and releases messengers that switch other cells on.

DEPENDS ON / TARGETS: Needs antigen-presenting cells (dendritic cells, macrophages). Activates B cells and CD8 killer cells.

Cytotoxic T cell (CD8)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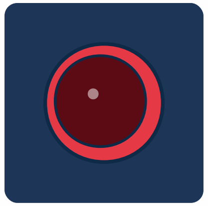
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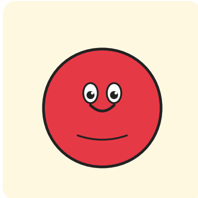
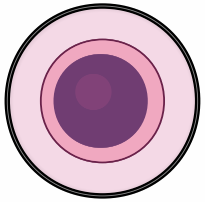

KIDS STYLE

FUNCTION: Killer cell: spots infected or tumour cells by their MHC-I signal and kills them precisely with perforin and granzymes.

DEPENDS ON / TARGETS: Primed by dendritic cells, often with help from CD4 T cells. Targets virus-infected body cells and tumour cells.

B cell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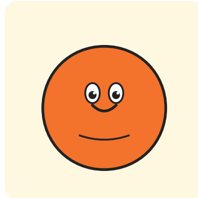
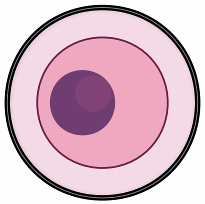

KIDS STYLE

FUNCTION: Antibody factory: binds antigen with its B-cell receptor and matures into plasma cells that pump out tailored antibodies.

DEPENDS ON / TARGETS: Usually needs help from CD4 T cells to switch into high gear. Targets free bacteria, virus particles and toxins in blood and tissue.

Plasma cell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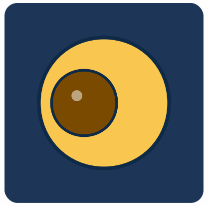
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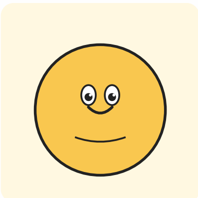
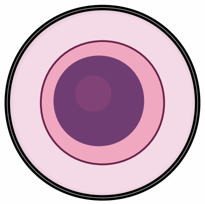

KIDS STYLE

FUNCTION: End stage of a B cell: a walking assembly line that secretes antibodies by the bucketload.

DEPENDS ON / TARGETS: Differentiates from an activated B cell. Its antibodies tag pathogens for other immune cells to deal with.

Natural killer cell (NK)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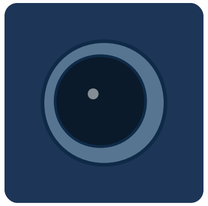
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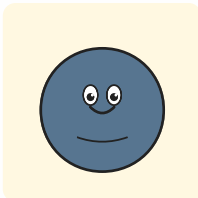
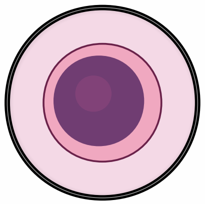

KIDS STYLE

FUNCTION: Innate-immune bouncer: kills cells that have lost their MHC-I 'ID badge' — typical for virus infection or cancer.

DEPENDS ON / TARGETS: Revved up by interferons and IL-12. Kills virus-infected and tumour cells without prior priming.

Regulatory T cell (Treg)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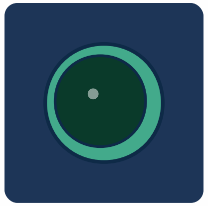
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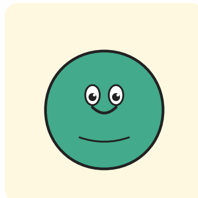

KIDS STYLE

FUNCTION: Brake pedal of the immune system: damps other immune cells so the response doesn't overshoot and turn into autoimmunity.

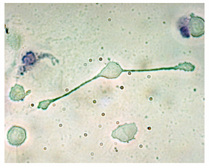
DEPENDS ON / TARGETS: Acts on helper T cells, CD8 killers and B cells — slows them down rather than killing them.

Immune cells — Phagocytes & Granulocytes

The rapid-response squad of innate immunity.

Phagocytes and granulocytes come from the myeloid lineage in bone marrow. They engulf, digest and neutralise invaders, patrolling blood, tissues and entry points like skin, lung and gut.

Macrophage



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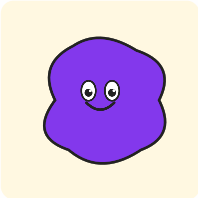
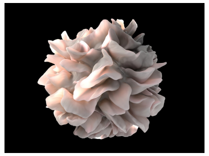


KIDS STYLE

FUNCTION: 'Big eater': engulfs bacteria, dead cells and debris, presents fragments to T cells and calls reinforcements with cytokines.

DEPENDS ON / TARGETS: Differentiates from blood monocytes. Phagocytoses bacteria, fungi, cancer cells and cleans up wounds.

Dendritic cell



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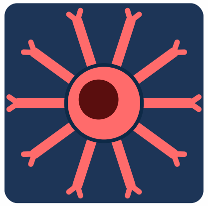

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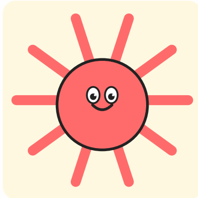
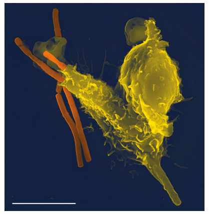


KIDS STYLE

FUNCTION: Scout and storyteller: gathers antigen in tissues, travels to lymph nodes and shows it to T cells — kicks off the adaptive response.

DEPENDS ON / TARGETS: Bridges innate and adaptive immunity. Activates helper T cells and CD8 killer cells.

Neutrophil



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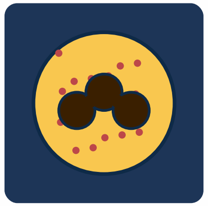

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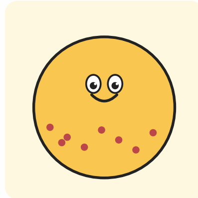
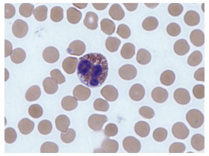


KIDS STYLE

FUNCTION: First responder: arrives first at sites of inflammation, eats bacteria, dumps antimicrobials and even casts sticky DNA nets (NETs).

DEPENDS ON / TARGETS: Chemokine-guided. Targets bacteria and fungi; lives only hours to a few days.

Eosinophil



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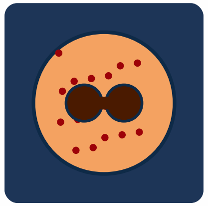

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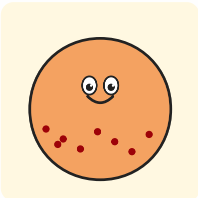
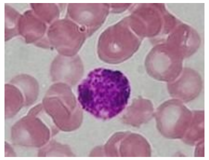


KIDS STYLE

FUNCTION: Specialist against parasitic worms and a player in allergic reactions. Packs big red granules full of toxic proteins.

DEPENDS ON / TARGETS: Recruited by IL-5 from Th2 and ILC2 cells. Targets parasites and amplifies asthma and allergic reactions.

Basophil



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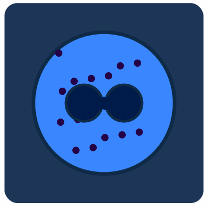

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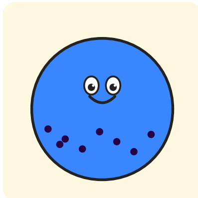
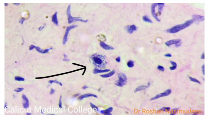


KIDS STYLE

FUNCTION: Rare in blood but loud: releases histamine and heparin and drives allergic and inflammatory reactions.

DEPENDS ON / TARGETS: Responds to IgE antibodies. Involved in allergies and in defence against parasites.

Mast cell



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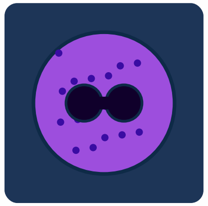

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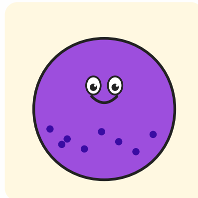


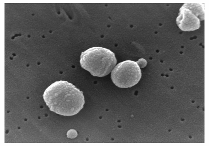
KIDS STYLE

FUNCTION: Tissue-resident sentinel packed with granules: on alarm it dumps histamine in an instant — swelling, itching, help signals.

DEPENDS ON / TARGETS: Triggered by IgE antibodies and pathogen signals. Drives allergy and anaphylaxis reactions.

Pathogens are anything that can cause disease: bacteria, viruses, fungi, parasites and prions. They live in air, water, soil, animals and on our skin — most microbes are harmless or beneficial, a few are highly dangerous.

Cocci (round bacteria)



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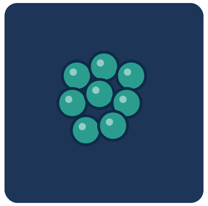

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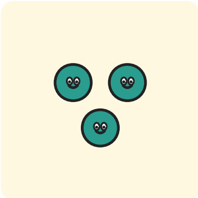
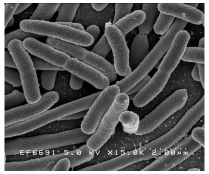


KIDS STYLE

FUNCTION: Round bacteria, often in clusters or chains (Staphylococci, Streptococci). Some are harmless guests, others cause pneumonia, skin or throat infections.

DEPENDS ON / TARGETS: Phagocytosed by neutrophils and macrophages; antibodies help tag them.

Rod bacteria



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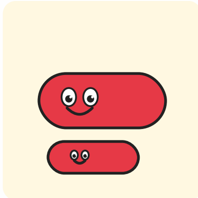
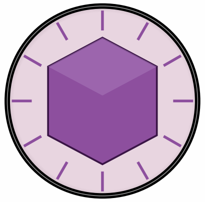


KIDS STYLE

FUNCTION: Elongated bacteria like E. coli, Salmonella, tuberculosis bacilli. Multiply fast and can release toxins.

DEPENDS ON / TARGETS: Phagocytosed; intracellular species (TB) need T-cell help to be cleared.

Virus



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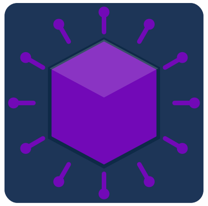

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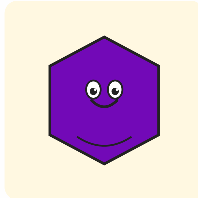


KIDS STYLE

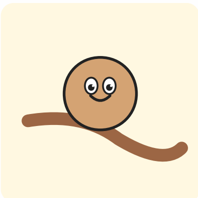
FUNCTION: Not really alive: just genetic material in a shell. Sneaks into body cells and forces them to build new virus particles.

DEPENDS ON / TARGETS: Hunted by CD8 T cells and NK cells; antibodies bind free virus particles.

Fungus

MICROSCOPE PHOTO

INFOGRAPHIC STYLE

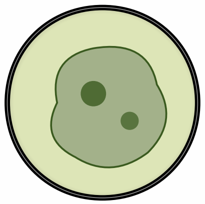


KIDS STYLE

FUNCTION: Eukaryotic microbes — yeasts or filaments (hyphae). Examples: Candida in mouth/gut, Aspergillus in lungs.

DEPENDS ON / TARGETS: Controlled by neutrophils, macrophages and Th17 cells.

Parasite



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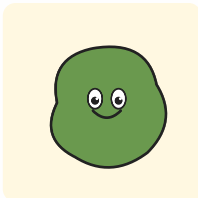
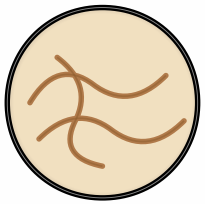


KIDS STYLE

FUNCTION: From single-celled (Plasmodium → malaria) to worms. Lives in or on its host and damages it in the process.

DEPENDS ON / TARGETS: Often fought with IgE, eosinophils and mast cells; antibodies and Th2 responses are central.

Prion



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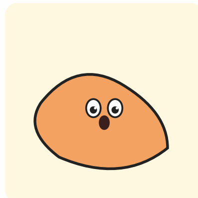


KIDS STYLE

FUNCTION: Not alive, not a virus — a misfolded protein that forces other proteins to misfold the same way. Causes diseases like Creutzfeldt-Jakob, BSE.

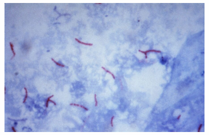
DEPENDS ON / TARGETS: Largely invisible to the immune system; no antibody defence.

Well-known bacteria

Famous representatives of the bacterial world.

Bacteria are single-celled organisms without a true nucleus (prokaryotes). They live everywhere — in soil and water, on skin and in the gut. Most are harmless or beneficial; a few cause millions of cases of disease worldwide every year.

Mycobacterium tuberculosis (TB)



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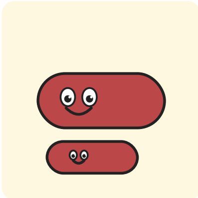
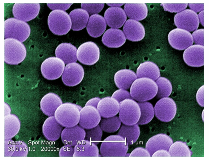


KIDS STYLE

FUNCTION: Waxy-coated rod that hides inside macrophages and can lurk in the lungs for years. Causes tuberculosis — still one of the deadliest infections worldwide.

DEPENDS ON / TARGETS: Airborne transmission. Contained by macrophages and CD4 T cells; without them the bacterium persists.

Staphylococcus aureus (MRSA)



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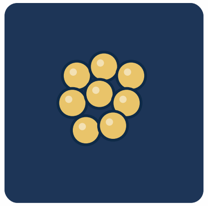

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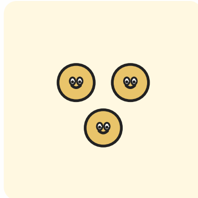
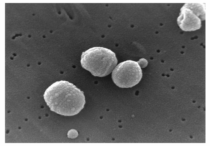


KIDS STYLE

FUNCTION: Grape-cluster cocci. Often harmless on skin and in the nose, but can cause wound infections, abscesses and sepsis. MRSA strains resist many antibiotics.

DEPENDS ON / TARGETS: Phagocytosed by neutrophils and tagged by antibodies. Hospital hygiene is key against resistance spread.

Streptococcus pneumoniae



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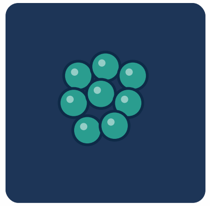

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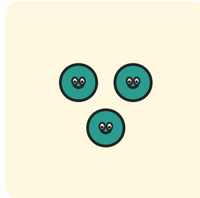
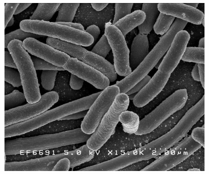


KIDS STYLE

FUNCTION: Cocci in chains, often in the nasopharynx. Leading cause of pneumonia, otitis media and meningitis, especially in children and the elderly.

DEPENDS ON / TARGETS: Polysaccharide capsule blocks phagocytosis; antibodies and complement break through. Vaccines available.

Escherichia coli



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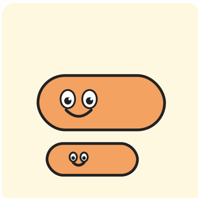
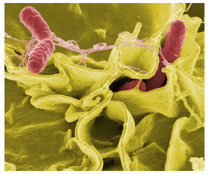


KIDS STYLE

FUNCTION: Gut rod bacterium. Most strains are harmless or useful; some (e.g. EHEC) produce toxins and cause severe diarrhoea and kidney failure.

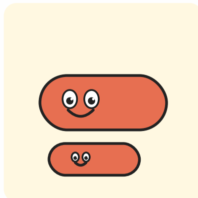
DEPENDS ON / TARGETS: Spread by faecal-oral route and contaminated food. Controlled in the gut by IgA antibodies and the mucus barrier.

Salmonella enterica



MICROSCOPE PHOTO

INFOGRAPHIC STYLE

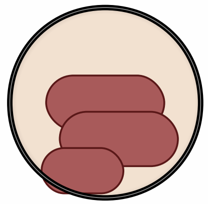


KIDS STYLE

FUNCTION: Rod bacterium from many animals' guts. Picked up via contaminated eggs, poultry or water, it causes diarrhoea (salmonellosis) or typhoid fever.

DEPENDS ON / TARGETS: Cleared by neutrophils, macrophages and the gut epithelium; Th1 responses are central to elimination.

Helicobacter pylori



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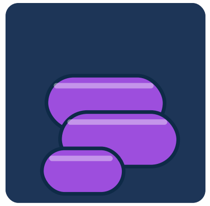

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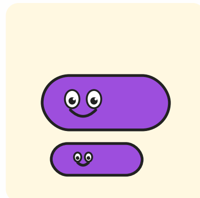


KIDS STYLE

FUNCTION: Spiral bacterium that survives in stomach acid. Causes gastritis, peptic ulcers and raises the risk of stomach cancer.

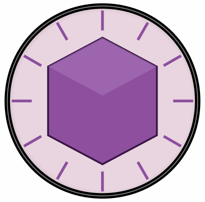
DEPENDS ON / TARGETS: Usually transmitted in childhood. Treated with antibiotics plus acid blockers; vaccine still in development.

Well-known viruses & other pathogens

Pandemic classics and global plagues.

Viruses are genetic material in a coat and need a host cell to replicate. Fungi and parasites, by contrast, are organisms in their own right. The pathogens shown here infect humans worldwide, ranging from harmless colds to severe pandemics.

Influenza virus (flu)



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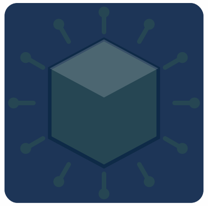


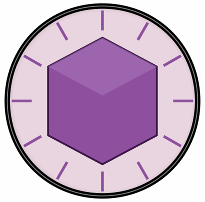
INFOGRAPHIC STYLE

KIDS STYLE

FUNCTION: Respiratory RNA virus with a segmented genome. Mutates constantly, so vaccines are reformulated each year. Causes seasonal waves and occasional pandemics.

DEPENDS ON / TARGETS: Spread by droplets. CD8 T cells and antibodies clear infected cells and free virus particles.

SARS-CoV-2 (COVID-19)



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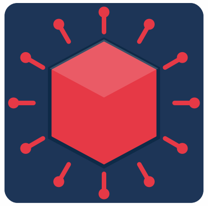

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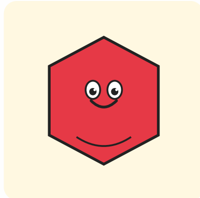
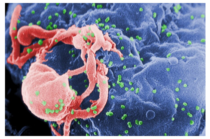


KIDS STYLE

FUNCTION: Coronavirus with a 'crown' of spike proteins. Binds ACE2 receptors in the airways and triggered the global COVID-19 pandemic in 2020.

DEPENDS ON / TARGETS: Spreads via aerosols. mRNA and vector vaccines train antibodies and T cells against the spike protein.

HIV



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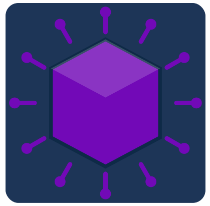

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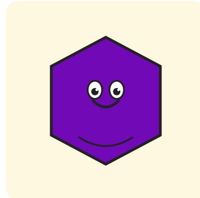
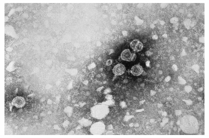


KIDS STYLE

FUNCTION: Retrovirus that targets CD4 helper T cells and integrates its genome into human DNA. Untreated, it destroys the immune system (AIDS).

DEPENDS ON / TARGETS: Spread via blood, sexual contact and mother-to-child. Antiretroviral therapy (ART) keeps the virus suppressed long-term.

Hepatitis B virus (HBV)



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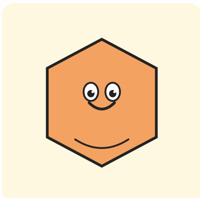
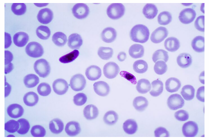


KIDS STYLE

FUNCTION: DNA virus that infects the liver and persists inside liver cells. Chronic infection can lead to liver cirrhosis and liver cancer.

DEPENDS ON / TARGETS: Spread via blood, sexual contact and perinatally. Vaccine available and included in many national immunisation programmes.

Plasmodium (malaria)



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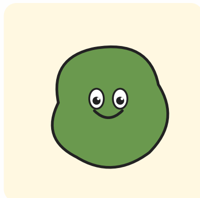


KIDS STYLE

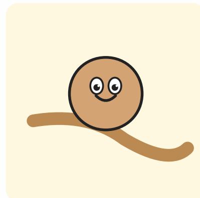
FUNCTION: Single-celled parasite spread by Anopheles mosquitoes. First infects the liver, then red blood cells, triggering cycles of fever. Kills over 500,000 people per year worldwide.

DEPENDS ON / TARGETS: Treated with antimalarial drugs; new vaccines (RTS,S, R21) now being rolled out in endemic regions.

Candida albicans

MICROSCOPE PHOTO

INFOGRAPHIC STYLE



KIDS STYLE

FUNCTION: Yeast that lives harmlessly in many people's mouths, gut and genital tract. With a weakened immune system or after antibiotics it can spread and cause serious infections.

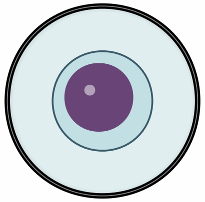
DEPENDS ON / TARGETS: Kept in check by Th17 cells, neutrophils and the mucosal microbiome.

Hematopoietic lineage

All blood cells arise from a single stem cell in bone marrow.

All blood cells arise in bone marrow from a single hematopoietic stem cell. Through multiple maturation steps they become red blood cells, platelets, granulocytes, monocytes and lymphocytes — around 200 billion new cells per person per day.

Hematopoietic stem cell (HSC)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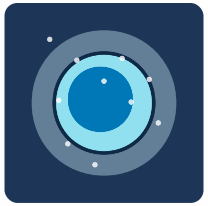
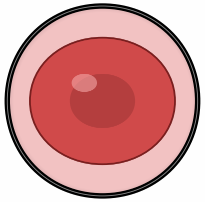

INFOGRAPHIC STYLE


KIDS STYLE

FUNCTION: Mother of all blood cells. Lives in bone marrow, divides rarely, self-renews and gives rise to every blood lineage.

DEPENDS ON / TARGETS: Needs niche cells in bone marrow (osteoblasts, stromal cells) and growth factors.

Erythrocyte (red blood cell)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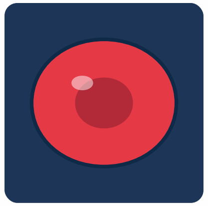
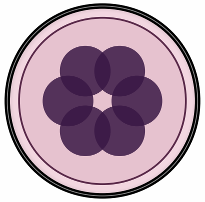

INFOGRAPHIC STYLE


KIDS STYLE

FUNCTION: Oxygen carrier: no nucleus, packed with haemoglobin. Lives about 120 days and circulates millions of times.

DEPENDS ON / TARGETS: Production driven by the hormone erythropoietin from the kidney. Delivers oxygen to every body cell.

Megakaryocyte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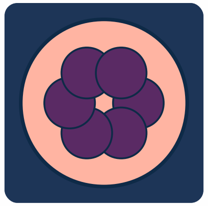
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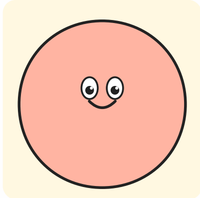
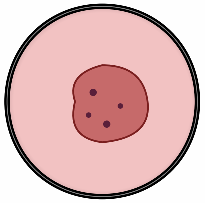

KIDS STYLE

FUNCTION: Giant bone-marrow cell with a multi-lobed nucleus. Pinches off thousands of platelets from its cytoplasm.

DEPENDS ON / TARGETS: Driven by thrombopoietin (TPO) from the liver. Provides the source material for platelets.

Thrombocyte (platelet)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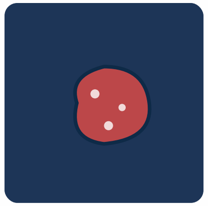
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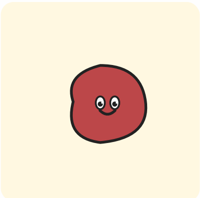
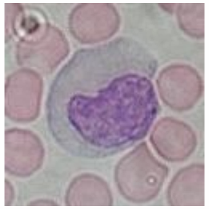

KIDS STYLE

FUNCTION: Tiny anucleate fragments. Plug wounds, form the first clot and trigger the coagulation cascade.

DEPENDS ON / TARGETS: Originate from megakaryocytes. Cooperate closely with coagulation factors and endothelial cells.

Monocyte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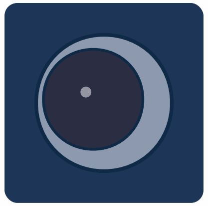
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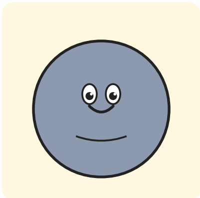
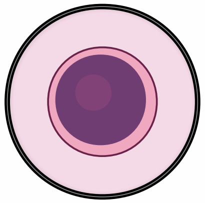

KIDS STYLE

FUNCTION: Large blood-borne precursor. Migrates into tissues and matures there into macrophages or dendritic cells.

DEPENDS ON / TARGETS: Responds to inflammatory signals. Supplies tissues with fresh phagocytes.

Lymphocyte (generic)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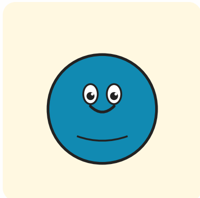
INFOGRAPHIC STYLE


KIDS STYLE

FUNCTION: Umbrella term for T, B and NK cells. Small, round, with a dense nucleus — the accountants of the immune system.

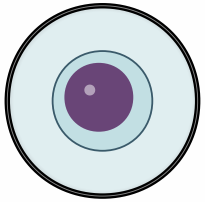
DEPENDS ON / TARGETS: Mature in bone marrow (B, NK) or thymus (T). Patrol lymph nodes, spleen and blood.

Mesenchymal lineage

Connective tissues: bone, cartilage, muscle, fat.

Mesenchymal cells arise from the middle germ layer (mesoderm) and form the body's supporting tissues: bone, cartilage, tendons, skeletal and smooth muscle, plus fat and connective tissue.

Mesenchymal stem cell (MSC)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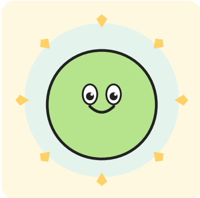
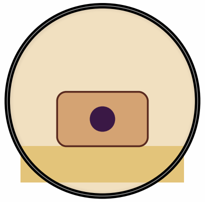

KIDS STYLE

FUNCTION: Reserve cell for supporting tissues. Can turn into bone, cartilage, muscle, fat or connective-tissue cells.

DEPENDS ON / TARGETS: Lives in bone marrow, fat tissue and umbilical cord. Supplies precursors on demand.

Osteoblast (bone builder)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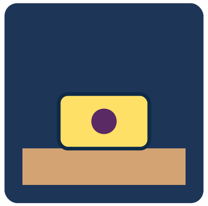
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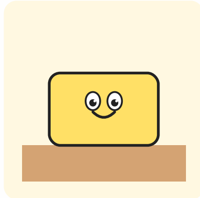
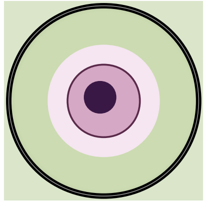

KIDS STYLE

FUNCTION: Builds new bone: secretes collagen and mineralises it with calcium. Later gets walled in as an osteocyte.

DEPENDS ON / TARGETS: Balances with osteoclasts that resorb bone. Hormonally controlled by PTH and vitamin D.

Chondrocyte (cartilage cell)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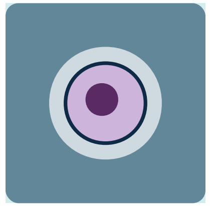
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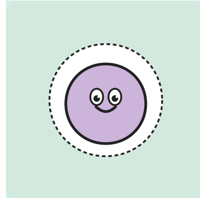
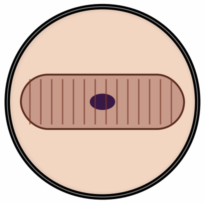

KIDS STYLE

FUNCTION: Sits alone or in small groups inside a 'lacuna' of cartilage matrix. Produces and maintains the rubbery cartilage tissue.

DEPENDS ON / TARGETS: Cartilage has almost no blood supply — cells rely on diffusion. Crucial for joints, intervertebral discs, growth plates.

Myocyte (muscle cell)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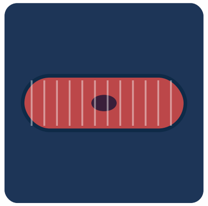
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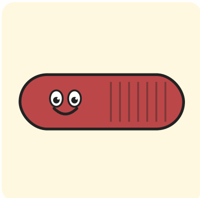
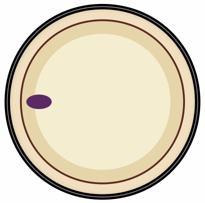

KIDS STYLE

FUNCTION: Long, fibrous cell packed with actin/myosin filaments. Shortens on an electrical signal and generates force and movement.

DEPENDS ON / TARGETS: Commanded by motor neurons, fuelled by glucose and oxygen from the blood.

Adipocyte (fat cell)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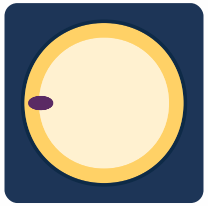
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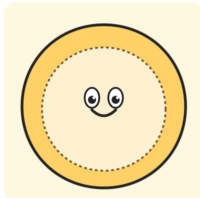
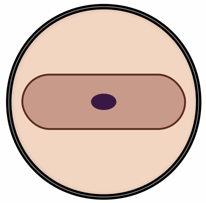

KIDS STYLE

FUNCTION: Lipid storage: a single huge fat droplet pushes the nucleus and cytoplasm to the rim. Also acts as an endocrine cell (e.g. leptin).

DEPENDS ON / TARGETS: Stimulated by insulin to take up fat. Releases energy for other cells when food is scarce.

Fibroblast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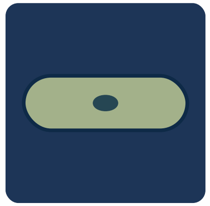
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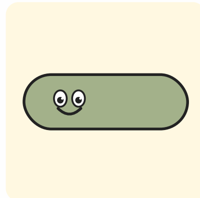
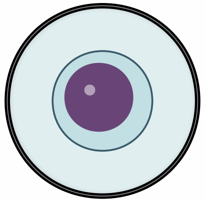

KIDS STYLE

FUNCTION: Connective-tissue all-rounder: makes collagen and other fibres, keeps skin and organs in shape and closes wounds.

DEPENDS ON / TARGETS: Activated at wounds by growth factors (TGF-β, PDGF). Works closely with macrophages.

Neurons and glia arise from the neural tube (ectoderm) and build the entire nervous system — brain, spinal cord and the peripheral nerves that supply muscles, skin and organs.

Neural stem cell (NSC)



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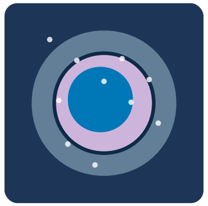

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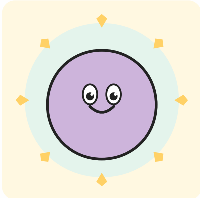
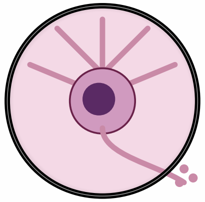


KIDS STYLE

FUNCTION: Precursor of all neural cell types. In the adult brain only active in a few niches (e.g. hippocampus).

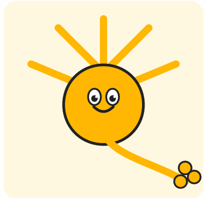
DEPENDS ON / TARGETS: Responds to growth factors (EGF, FGF) and to tissue cues.

Neuron



MICROSCOPE STYLE

INFOGRAPHIC STYLE

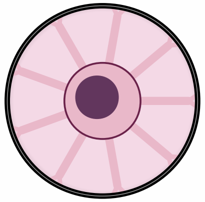


KIDS STYLE

FUNCTION: Compute-and-signal cell: collects inputs through dendrites, sums them up and fires electrical impulses down its axon.

DEPENDS ON / TARGETS: Depends on astrocytes (support) and oligodendrocytes (myelin). Communicates with other neurons through synapses.

Astrocyte



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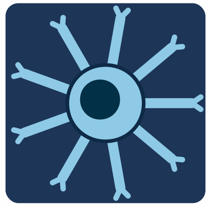

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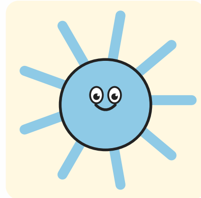
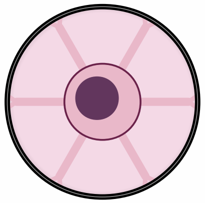


KIDS STYLE

FUNCTION: Star-shaped support cell. Feeds neurons, balances ions and neurotransmitters, helps form parts of the blood-brain barrier.

DEPENDS ON / TARGETS: Supports neurons and regulates local blood flow.

Oligodendrocyte



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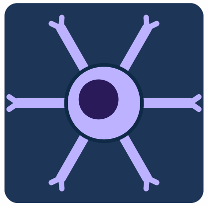

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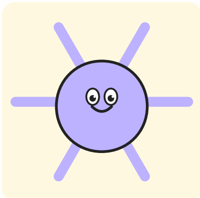
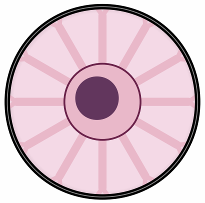


KIDS STYLE

FUNCTION: Wraps several axons at once with myelin in brain and spinal cord — the electrical insulation of nerves.

DEPENDS ON / TARGETS: Partly fuels axons. In multiple sclerosis this myelin layer is the target.

Microglia



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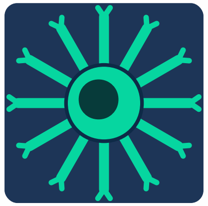

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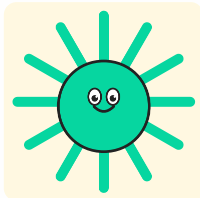
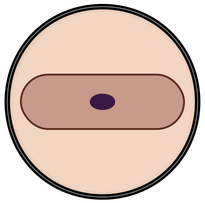


KIDS STYLE

FUNCTION: Brain-resident macrophages: constantly probe the tissue with thin processes, clear debris, prune synapses.

DEPENDS ON / TARGETS: Yolk-sac derived in the embryo. First line of defence against brain infections.

Schwann cell



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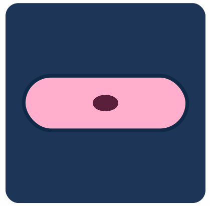

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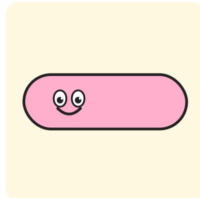


KIDS STYLE

FUNCTION: Myelin maker in the peripheral nervous system: each Schwann cell wraps one segment of a single axon.

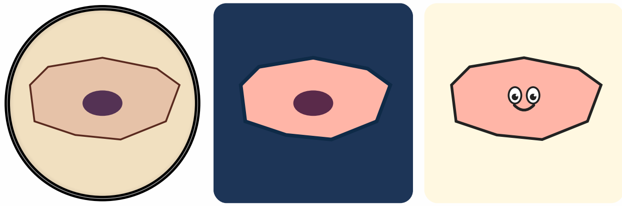
DEPENDS ON / TARGETS: Also helps peripheral nerves regenerate after injury.

Epithelial lineage

Body surfaces: skin, gut, lung.

Epithelial cells line every inner and outer surface of the body: skin, the mucosa of mouth, GI tract and lung, plus glands. They form the body's first mechanical and biochemical barrier against the environment.

Keratinocyte (skin cell)



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FUNCTION: Building block of the outer skin. Lays down keratin, cornifies upward and forms a waterproof, robust barrier.

DEPENDS ON / TARGETS: Renewed from stem cells in the basal layer. Cooperates with melanocytes and Langerhans cells.

Enterocyte (gut cell)

MICROSCOPE STYLE | INFOGRAPHIC STYLE | KIDS STYLE

FUNCTION: Tall column-shaped cell with a dense brush border (microvilli). Absorbs nutrients from the gut and passes them on.

DEPENDS ON / TARGETS: Replaced every few days by intestinal stem cells. Forms a barrier against gut microbes.

Goblet cell

MICROSCOPE STYLE | INFOGRAPHIC STYLE | KIDS STYLE

FUNCTION: Goblet-shaped mucus producer in gut and airways. Releases mucin packets that form a protective gel over the epithelium.

DEPENDS ON / TARGETS: Cooperates with enterocytes. Protects against dryness and bacterial contact.

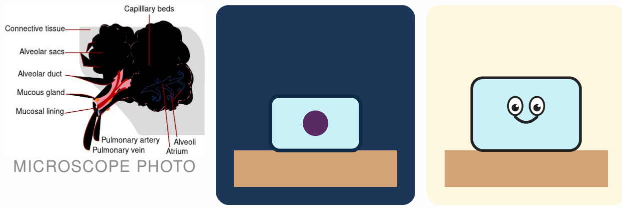
Paneth cell

MICROSCOPE STYLE | INFOGRAPHIC STYLE | KIDS STYLE

FUNCTION: Sits at the base of intestinal crypts and fires antimicrobial proteins (defensins, lysozyme). Protects the intestinal stem cells.

DEPENDS ON / TARGETS: Together with stem cells forms the crypt-base niche.

Alveolar cell type II



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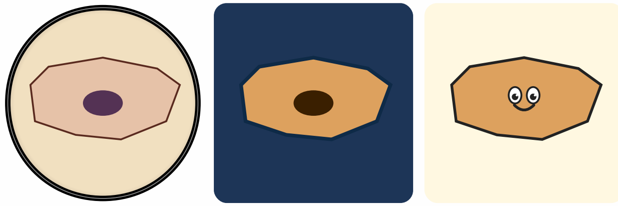
FUNCTION: Small cuboidal lung cell. Makes surfactant — the film that keeps alveoli open. Also acts as stem cell for type I cells.

DEPENDS ON / TARGETS: Lives among the thin type-I cells that handle gas exchange.

Urothelial cell



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FUNCTION: Stretchable cell of the bladder wall. Reshapes its surface as the bladder fills and seals against acidic urine.

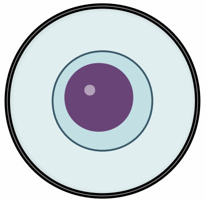
DEPENDS ON / TARGETS: Forms a barrier between urine and body. Renewed from basal cells.

Endothelial lineage

Inner lining of every blood and lymph vessel.

Endothelial cells form a single layer that lines the inside of every blood and lymph vessel, from heart to the smallest capillary. They regulate blood flow, clotting and the passage of immune cells out of the bloodstream into tissues.

Endothelial progenitor (EPC)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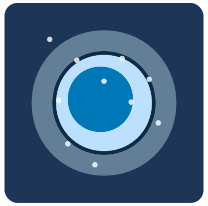
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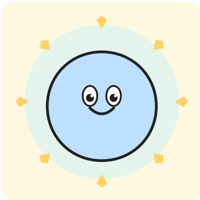
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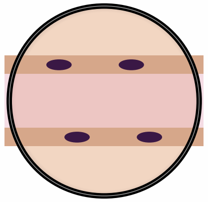
FUNCTION: Young endothelial cell. Mobilised during wound healing and angiogenesis to replace damaged vessel wall.

DEPENDS ON / TARGETS: Recruited from bone marrow; responds to VEGF.

Endothelial cell (continuous)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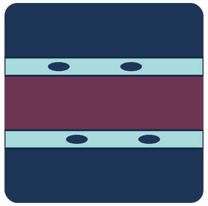
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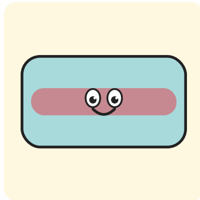
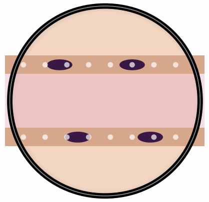

KIDS STYLE

FUNCTION: Flat cobblestone that lines most vessels. Regulates what diffuses between blood and tissue.

DEPENDS ON / TARGETS: Responds to blood-flow shear and hormones (e.g. histamine).

Fenestrated endothelium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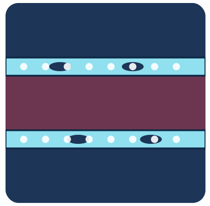
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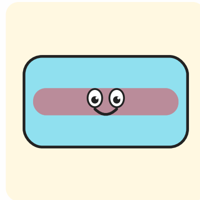
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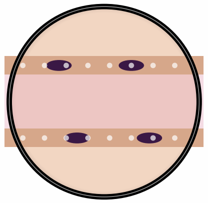
FUNCTION: Endothelium with little 'windows' — permeable to fluid and molecules. Found in kidney, gut and endocrine glands.

DEPENDS ON / TARGETS: Filters blood, releases hormones quickly or takes up nutrients.

Sinusoidal endothelium (liver)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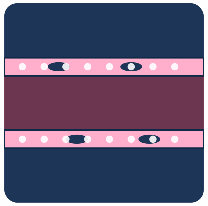
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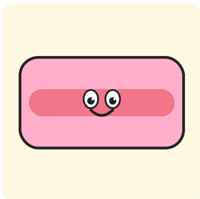
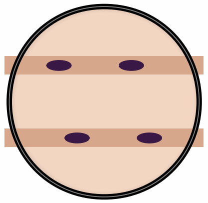

KIDS STYLE

FUNCTION: Very gappy — even large molecules pass through. Allows the liver direct contact with bloodflow.

DEPENDS ON / TARGETS: Works closely with hepatocytes and Kupffer cells (liver macrophages).

Lymphatic endothelium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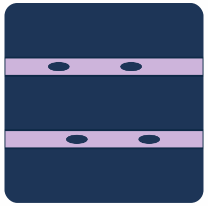
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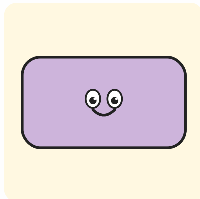
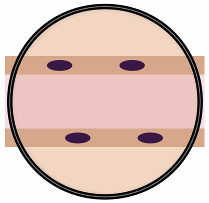

KIDS STYLE

FUNCTION: Wall of lymphatic vessels. The cells overlap like flaps so lymph flows only one way.

DEPENDS ON / TARGETS: Collects excess tissue fluid and returns it to the bloodstream.

Blood-brain-barrier endothelium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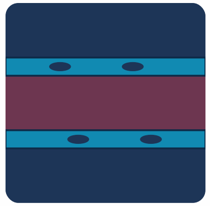
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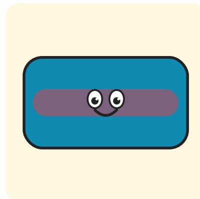

KIDS STYLE

FUNCTION: Extra-tight: cells are locked together by tight junctions. Keeps almost everything out of the brain.

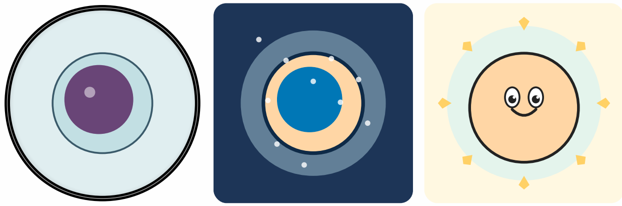
DEPENDS ON / TARGETS: Cooperates with astrocytes and pericytes that help form the barrier.

Induced pluripotent stem cells (iPS)

Lab-reprogrammed body cells — source for many therapies.

Induced pluripotent stem cells are made in the lab: body cells (e.g. skin cells) are reprogrammed with defined factors back into an embryonic-like stem-cell state and can then differentiate into almost any cell type.

iPS cell

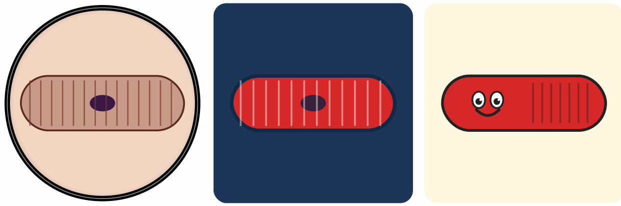


MICROSCOPE STYLE | INFOGRAPHIC STYLE | KIDS STYLE

FUNCTION: A body cell (e.g. skin) reset by a handful of transcription factors into an embryonic-like state — can develop into almost any cell type.

DEPENDS ON / TARGETS: Differentiated in vitro into a target lineage with specific growth factors.

Cardiomyocyte (heart muscle cell)



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FUNCTION: Striated muscle cell that beats rhythmically. Connected by intercalated discs so the heartbeat stays in sync.

DEPENDS ON / TARGETS: Responds to signals from the sinoatrial node and to hormones (adrenaline).

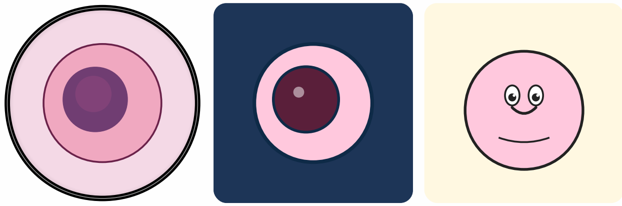
Hepatocyte (liver cell)

MICROSCOPE STYLE | INFOGRAPHIC STYLE | KIDS STYLE

FUNCTION: Metabolic all-rounder: breaks down toxins, makes bile and proteins, stores glucose as glycogen.

DEPENDS ON / TARGETS: Receives blood from portal vein (gut) and hepatic artery. Can divide and regenerate after damage.

Beta cell (pancreas)



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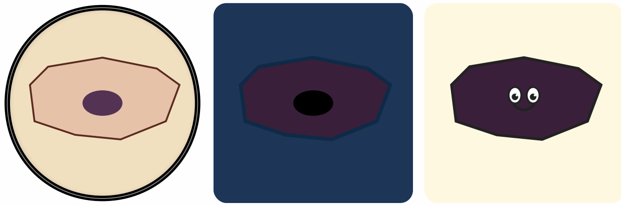
FUNCTION: Insulin maker in the islets of Langerhans. Senses blood glucose and releases insulin when it rises.

DEPENDS ON / TARGETS: Destroyed by the immune system in type-1 diabetes — replacing it from iPS cells is an active research area.

Retinal pigment epithelium (RPE)



MICROSCOPE STYLE | INFOGRAPHIC STYLE | KIDS STYLE

FUNCTION: Dark cell layer behind the photoreceptors. Recycles spent receptor tips and supplies the photoreceptors with nutrients.

DEPENDS ON / TARGETS: Fails in age-related macular degeneration — iPS-derived RPE cells are being tested as transplants.

Dopaminergic neuron

MICROSCOPE STYLE | INFOGRAPHIC STYLE | KIDS STYLE

FUNCTION: Neuron that releases dopamine. Crucial for reward, motivation and movement planning.

DEPENDS ON / TARGETS: Dies off in Parkinson's disease — iPS-derived dopamine neurons are being explored as cell-replacement therapy.

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